AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Stonewall Creek Business Park, VA -- station KJYO, America/New_York
Committed pair	OSM 1443922527 -> 1460776171 OSM way 1443922527 / OSM way 1460776171
Facade gap	112.4 m between the two halls
Weather record	43,249 real hourly records from KJYO
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3496 C at 305 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-04-08 (safe_sector)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 2 of 24 hours with 1 mode change. The reactive incumbent took 8 hours -- 6 more -- but declared 0 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 7 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	2 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	8 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 7 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dry-bulb	18.300	18.0	18.000	1.180
01	mechanical	no	dry-bulb	18.066	18.0	17.208	1.124
02	mechanical	yes	switch budget	17.196	18.0	16.000	0.843
03	mechanical	yes	switch budget	17.299	18.0	16.000	0.893
04	mechanical	yes	switch budget	17.066	18.0	15.208	1.396

05	mechanical	yes	switch budget	15.739	18.0	14.000	1.207
06	mechanical	yes	switch budget	15.566	18.0	13.208	1.174
07	mechanical	yes	switch budget	15.734	18.0	13.000	1.512
08	mechanical	yes	switch budget	16.712	18.0	14.000	1.565
09	mechanical	no	dry-bulb	18.694	18.0	16.000	2.227
10	mechanical	no	dry-bulb	19.638	18.0	17.000	2.349
11	mechanical	no	dry-bulb	20.172	18.0	18.000	2.099
12	mechanical	no	dry-bulb	21.722	18.0	21.000	1.491
13	mechanical	no	dry-bulb	21.613	18.0	21.000	1.252
14	mechanical	no	dry-bulb	21.679	18.0	21.000	1.190
15	mechanical	no	dry-bulb	23.186	18.0	22.000	1.134
16	mechanical	no	dry-bulb	22.754	18.0	22.000	1.206
17	mechanical	no	dry-bulb	21.642	18.0	21.000	1.129
18	mechanical	no	dry-bulb	21.253	18.0	20.000	1.206
19	mechanical	no	dry-bulb	20.199	18.0	18.000	1.532
20	mechanical	no	dry-bulb	19.307	18.0	17.000	1.827
21	mechanical	no	dry-bulb	18.253	18.0	16.000	1.754
22	FREE	yes	-	16.671	18.0	15.000	1.461
23	FREE	yes	-	16.158	18.0	15.000	1.235

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.300 C against a 18.0 C limit, so it fails by 0.300 C. It would take a limit 0.300 C higher, or a bound 0.300 C tighter, to change this hour.

01:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.066 C against a 18.0 C limit, so it fails by 0.066 C. It would take a limit 0.066 C higher, or a bound 0.066 C tighter, to change this hour.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.196 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.299 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.066 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.739 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.566 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.734 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.712 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 18.694 C against a 18.0 C limit, so it fails by 0.694 C. It would take a limit 0.694 C higher, or a bound 0.694 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 19.638 C against a 18.0 C limit, so it fails by 1.638 C. It would take a limit 1.638 C higher, or a bound 1.638 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 20.172 C against a 18.0 C limit, so it fails by 2.172 C. It would take a limit 2.172 C higher, or a bound 2.172 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.722 C against a 18.0 C limit, so it fails by 3.722 C. It would take a limit 3.722 C higher, or a bound 3.722 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.613 C against a 18.0 C limit, so it fails by 3.613 C. It would take a limit 3.613 C higher, or a bound 3.613 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.679 C against a 18.0 C limit, so it fails by 3.679 C. It would take a limit 3.679 C higher, or a bound 3.679 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 23.186 C against a 18.0 C limit, so it fails by 5.186 C. It would take a limit 5.186 C higher, or a bound 5.186 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 22.754 C against a 18.0 C limit, so it fails by 4.754 C. It would take a limit 4.754 C higher, or a bound 4.754 C tighter, to

change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.642 C against a 18.0 C limit, so it fails by 3.642 C. It would take a limit 3.642 C higher, or a bound 3.642 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.253 C against a 18.0 C limit, so it fails by 3.253 C. It would take a limit 3.253 C higher, or a bound 3.253 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.199 C against a 18.0 C limit, so it fails by 2.199 C. It would take a limit 2.199 C higher, or a bound 2.199 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.307 C against a 18.0 C limit, so it fails by 1.307 C. It would take a limit 1.307 C higher, or a bound 1.307 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.253 C against a 18.0 C limit, so it fails by 0.253 C. It would take a limit 0.253 C higher, or a bound 0.253 C tighter, to change this hour.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.671 C, which is 1.329 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.461 C, and the margin is measured, not chosen: 1.290 C of group-conditional forecast error for this hour of day, plus 0.1712 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.158 C, which is 1.842 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.235 C, and the margin is measured, not chosen: 1.078 C of group-conditional forecast error for this hour of day, plus 0.1578 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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